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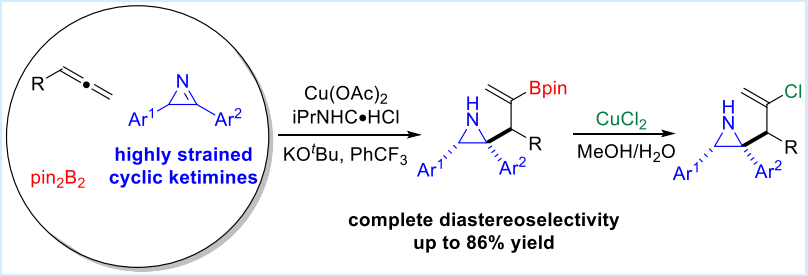
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Copper-Catalyzed Three-Component Carboboronation of Allenes Using Highly Strained Cyclic Ketimines as Electrophiles

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Cite This: *Org. Lett.* 2021, 23, 4431−4435 Read Online

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ABSTRACT: A diastereoselective copper and NHC-ligand-catalyzed three-component difunctionalization of allenes with bis(pinacolato)diboron and 2*H*-azirines to afford borylated allylaziridines is described. The reaction exhibits complete diastereoselectivity and good yields, and the further chlorination of the corresponding borylated products was also performed. It is believed that the high ring-strain force of 2*H*-azirines facilitates the reaction. More chemical transformations of borylated allylaziridines are also reported.

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Boron-containing compounds play important roles in organic synthesis,[1a,b] catalysis,[1c−e] materials,[1f,g] and therapeutic drug discovery.[1h] Therefore, the development of efficient methods to obtain such compounds is highly desirable. Among various methods to acquire them, the transition-metal-catalyzed borylative difunctionalization of unsaturated hydrocarbons is highly attractive because it usually affords multifunctional boron-containing products, which are favorable building blocks in organic synthesis.[2] Copper has been found to be the most attractive catalyst for such reactions.[3] In a typical reaction pathway, the boron-containing organocopper species is formed in situ via the addition of borylcopper to alkenes, alkynes, or allenes, which would then undergo nucleophilic additions or substitutions with electrophiles.[3e] However, unlike organocuprates (organocopperate complex),[4] the in situ-generated boron-substituted organocopper species exhibit a unique nucleophilic reactivity since only specific electrophiles are suitable substrates. For example, the allylcopper species[3a−c] could undergo nucleophilic reactions with a few types of halides,[5] allyl esters,[6] ketones, aldehydes,[7] imines,[8] highly electron-deficient alkenes,[9] and other electrophiles.[10]

When imines are used as the electrophiles, the corresponding products of their reaction with the allylcopper species are boron-substituted homoallylic amines, which are important intermediates to synthesize natural products and bioactive molecules. Therefore, reactions with allylcopper species that employ imines as electrophiles attract considerable attention (Scheme 1).[11] Aldimines were first used as electrophilic reaction partners in the three-component carboboronation of allenes with bis(pinacolato)diboron (B_2Pin_2) by Procter.[8a,b] However, ketimines have proven to be much more challenging substrates than aldimines for the reaction until Hoveyda made a breakthrough by utilizing unprotected ketimine salts as the electrophiles, and the nitrogen atom bearing a positive charge makes the ketimine salts more prone to an attack by the nucleophilic organocopper species.[8g] Very recently, Procter reported that ketiminoesters are also favorable substrates for the boron-containing allylcopper species because the electron-withdrawing ester group activates the ketimines.[8c]

Our group is interested in employing cyclic imines as electrophiles in the copper-catalyzed carboboronation of allenes and has developed a copper-catalyzed enantioselective carboboronation reaction of allenes using seven-membered cyclic aldimines as electrophiles.[12] However, when seven-membered cyclic ketimines were subjected to the reaction, no desired product was detected. We then wondered whether highly strained three-membered cyclic ketimines could be

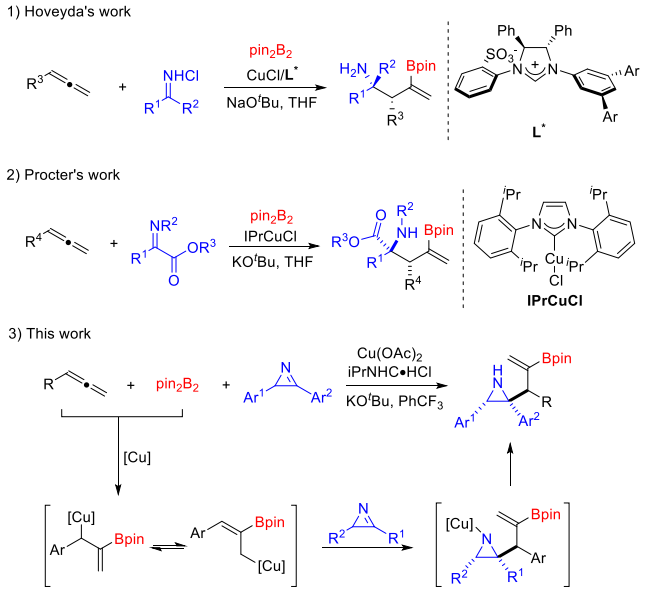
Received: April 21, 2021
Published: May 13, 2021



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https://doi.org/10.1021/acs.orglett.1c01372
Org. Lett. 2021, 23, 4431−4435

Scheme 1. Nucleophilic Addition of In Situ-Generated Allylcopper Species to Ketimines

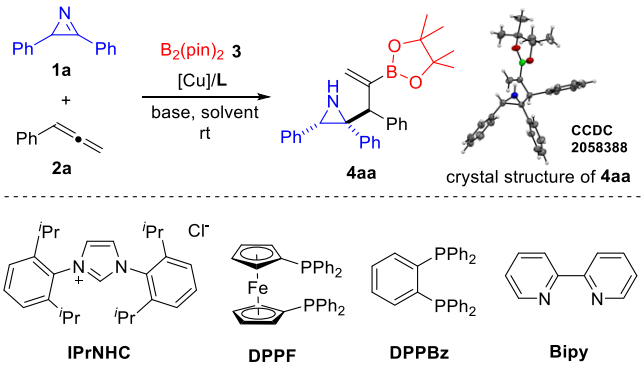


active enough to be attacked by the in situ-formed boron-substituted allylcopper species.[13] As a result, we report here a copper-catalyzed carboboronation of allenes with diboronates and 2*H*-azirines.

Initially, we selected the reaction of 2,3-diphenyl-2*H*-azirine **1a**, phenyl allene **2a**, and bis(pinacolato)diboron **3** as the model reaction. Then, we probed a series of copper salts, ligands, bases, solvents, temperatures, and loading ratios of substrates to obtain the optimal reaction conditions (Table 1). To our delight, the desired product **4aa** was obtained in a 76% yield and >20:1 dr in the presence of 10 mol % CuCl and 15 mol % IPrNHC using ^{t}BuOK as the base and $PhCF_3$ as the solvent at room temperature (Table 1, entry 1), and the crystal structure of product **4aa** was unambiguously identified. However, an evaluation of other ligands (such as DPPF, DPPBz, bipy, et al.) indicated that *P*,*P*- and *N*,*N*-ligands impaired the reaction yield remarkably, and IPrNHC was still the best ligand (Table 1, entries 2−4, respectively). Switching ^{t}BuOK to other bases, such as ^{t}BuONa, ^{t}BuOLi, and MeOK, resulted in lower yields of **4aa** (Table 1, entries 5−7, respectively). We then screened several other copper salts, including CuBr, CuI, $Cu(OAc)_2$, $Cu(OTf)_2$, and $CuBr_2$ (Table 1, entries 8−12, resptively). Among them, $Cu(OAc)_2$ exhibited the best performance, and product **4aa** was obtained in an 83% yield (Table 1, entry 10). Examinations of different solvents indicated that $PhCF_3$ was still the best solvent (Table 1, entries 13−15). Both elevating and decreasing the reaction temperature was harmful to the reaction, since lower yields of **4aa** were obtained (Table 1, entries 16 and 17). The loading ratio of the copper salt to the ligand had a huge effect on the reaction, since the yield of **4aa** was greatly decreased when a 1:1 ratio of $Cu(OAc)_2$/IPrNHC was used as the catalyst (Table 1, entry 18).

After the optimal conditions were obtained, we set out to evaluate the performance of various substituted 2*H*-azirine **1** and allenes **2** for the reaction. It was found that most of the boronated products **4** were difficult to be characterized by NMR (nuclear magnetic resonance) spectroscopy due to bad signal splitting in the ^{1}H NMR spectrum and missing signals in

Table 1. Optimization of Reaction Conditions[a]



entry	[M]	L	base	solvent	yield (%)[b,c]
1	CuCl	IPrNHC	^{t}BuOK	$PhCF_3$	76
2	CuCl	DPPF	^{t}BuOK	$PhCF_3$	58
3	CuCl	DPPBz	^{t}BuOK	$PhCF_3$	65
4	CuCl	Bipy	^{t}BuOK	$PhCF_3$	40
5	CuCl	IPrNHC	^{t}BuONa	$PhCF_3$	74
6	CuCl	IPrNHC	^{t}BuOLi	$PhCF_3$	69
7	CuCl	IPrNHC	MeOK	$PhCF_3$	57
8	CuBr	IPrNHC	^{t}BuOK	$PhCF_3$	61
9	CuI	IPrNHC	^{t}BuOK	$PhCF_3$	66
10	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	$PhCF_3$	83
11	$Cu(OTf)_2$	IPrNHC	^{t}BuOK	$PhCF_3$	61
12	$CuBr_2$	IPrNHC	^{t}BuOK	$PhCF_3$	57
13	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	toluene	80
14	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	THF	76
15	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	1,4-dioxane	72
16[d]	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	$PhCF_3$	49
17[e]	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	$PhCF_3$	73
18[f]	$Cu(OAc)_2$	IPrNHC	^{t}BuOK	$PhCF_3$	53

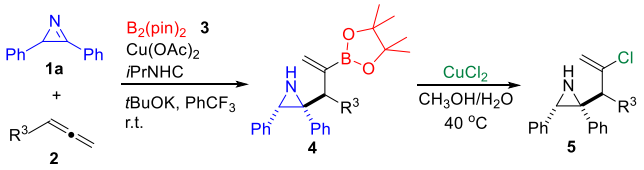
[a]All reactions were carried out with **1a** (0.2 mmol), **2a** (0.6 mmol), **3** (0.4 mmol), KO^{t}Bu (0.2 mmol), copper salt (0.02 mmol), and ligand (0.03 mmol) in 3 mL of solvent at 25 °C for 12 h. [b]Total isolated yield. [c]For all the reactions, dr >20:1, and the values were determined by a ^{1}H NMR analysis. [d]The reaction was carried out at 0 °C. [e]The reaction was carried out at 60 °C. [f]The ligand loading was 0.02 mmol.

the ^{13}C NMR spectrum.[14] Therefore, all of them were then converted to chlorinated products **5** using $CuCl_2$ as the chlorine source for a better NMR analysis, and the yields of **4** and total yields of **5** in two steps are reported in Tables 2 and 3, respectively. Complete diastereoselectivity was obtained for all the reactions.

For all the allene substrates, the corresponding products were obtained with moderate to good yields. For arylallenes, alkyl-substituted phenylallenes (Table 2, entries 2−4) afforded the corresponding products with lower yields than those from halogen- or oxygen-substituted phenylallenes (Table 2, entries 5−7, respectively). Noticeably, the CF_3-substituted phenylallene **2h** afforded the corresponding product **5ah** in a considerablely low yield (Table 2, entry 8). Additionally, allenes with substituents at different positions of the phenyl ring exhibited similar reactivities under the optimal conditions, and the desired products were obtained in 68%−75% total yields (Table 2, entries 9−12). However, alkylallene **5m** exhibited a much weaker reactivity than those of the phenyl allenes , and the corresponding boronated product was obtained in a 36% yield. (Table 2, entry 13). To further expand the scope of the allene substrates, other types of

https://doi.org/10.1021/acs.orglett.1c01372
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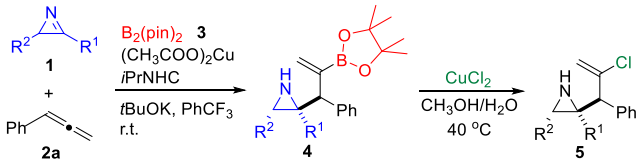
Table 2. Scope of Allenes[a]



entry	R^3	5	yield of 4 (%)[b]	yield of 5 (%)[c]
1	C_6H_5	5aa	83	78
2	4-MeC_6H_4	5ab	67	60
3	4-EtC_6H_4	5ac	65	62
4	4-^{t}BuC_6H_4	5ad	72	65
5	4-ClC_6H_4	5ae	75	72
6	4-BrC_6H_4	5af	80	75
7	4-$MeOC_6H_4$	5ag	83	71
8	4-$CF_3C_6H_4$	5ah	63	55
9	3-MeC_6H_4	5ai	73	68
10	3-ClC_6H_4	5aj	80	71
11	2-ClC_6H_4	5ak	86	75
12	3,4-$Me_2C_6H_3$	5al	76	69
13	cyclohexyl	5am	36	24

[a]All reactions were carried out with **1a** (0.2 mmol), **2** (0.6 mmol), **3** (0.4 mmol), and ^{t}BuOK (0.2 mmol) with $Cu(OAc)_2$ (10 mol %) and ^{i}PrNHC(15 mmol %) in 3 mL of $PhCF_3$ at 25 °C for 12h. [b]Isolated yield. [c]Total isolated yield.

Table 3. Scope of 2*H*-Azirines[a]



entry	R^1	R^2	5	yield of 4 (%)[b]	yield of 5(%)[c]
1	4-MeC_6H_4	C_6H_5	5ba	80	75
2	4-^{n}PrC_6H_4	C_6H_5	5ca	83	76
3	4-ClC_6H_4	C_6H_5	5da	81	75
4	C_6H_5	4-BrC_6H_4	5ea	82	79
5	C_6H_5	4-MeC_6H_4	5fa	80	72
6	C_6H_5	4-ClC_6H_4	5ga	78	72
7	C_6H_5	2-ClC_6H_4	5ha	79	75
8	C_6H_5	2-MeC_6H_4	5ia	83	75
9	C_6H_5	3-MeC_6H_4	5ja	86	78
10	C_6H_5	3-ClC_6H_4	5ka	85	78
11	C_6H_5	H	5la		

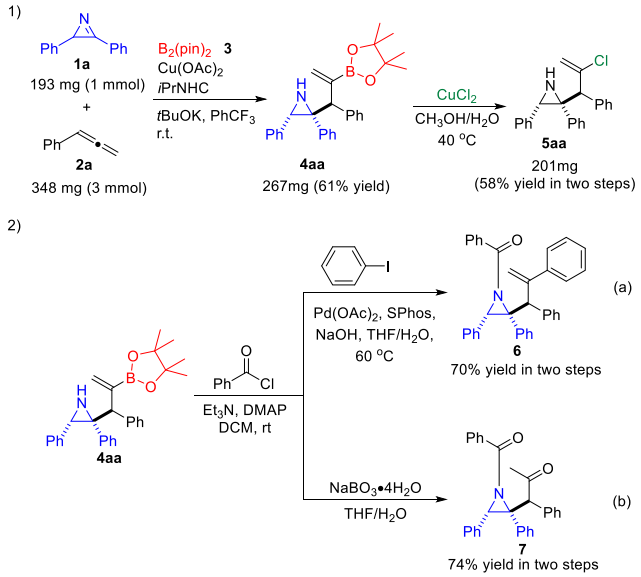
[a]Unless otherwise indicated, all reactions were carried out with **1** (0.2 mmol), **2a** (0.6 mmol), **3** (0.4 mmol), and *t*BuOK (0.2 mmol) with $Cu(OAc)_2$ (10 mol %) and ^{i}PrNHC (15 mmol %) in 3 mL of $PhCF_3$ at 25 °C for 12h. [b]Isolated yield. [c]Total isolated yield.

allenes, such as α-substituted phenylallene **2n**, phenoxylallene **2o**, and allenamide **2p**, were also subjected to the optimal reaction conditions. Unfortunately, none of them were suitable substrates for the reaction since no desired products was detected.

The substrate scope of 2*H*-Azirines was then evaluated, and substrates with variations on both R^1 and R^2 groups were examined under the standard conditions. When R^1 and R^2 were substituted phenyl groups, little influence on product yields was observed. When R^2 was a phenyl group, the substrates bearing substituents with different electronic properties (Me, *n*Pr, and Cl) on the *para*-position of phenyl rings afforded good yields of the desired products (Table 3, entries 1−3, respectively). Similarly, When R^1 was a phenyl group, the corresponding products were obtained in high yields regardless of the presence of alkyl or halogen substituents or their substituted positions on phenyl rings of R^2. However, when R^1 was H, no desired product was obtained, and we found that the substrate decomposed rapidly under standard conditions.

Followed by the expansion of substrate scope, a scaled-up synthesis of **4aa** was then evaluated. When the reaction of **1a** was scaled up to 1 mmol (based on **1a**) under the optimal conditions, **4aa** was obtained in a 61% yield, which is considerably lower than that in a smaller-scale synthesis. However, the yield for the chlorination reaction remained almost the same (Scheme 2.1). In addition to the chlorination

Scheme 2. Scaled-Up Synthesis and Derivation of Borylated Product 4aa



of the boryl group in product **4**, more transformations of the boryl group were tried. The protection of the amine group can be achieved by reacting compound **4a** with benzoyl chloride in the presence of Et_3N and DMAP using DCM as the solvent, and the protected product **4aa** underwent a paladium-catalyzed cross-coupling reaction with iodobenzene in a 70% yield over two steps (Scheme 2.2a). The protected product could also be converted to the corresponding ketone **7** in a 74% yield in two steps via a typical oxidative workup process (Scheme 2.2b) (Scheme 2(1)). The asymmetric version of the synthesis of compound **4aa** was also investigated. However, only a poor yield and a moderate enantioselecvitity were obtained.[15]

In summary, we have developed a copper-catalyzed three-component carboboronation of arylallenes using 2*H*-azirines as the electrophiles. It shows that such highly strained three-membered ketimines with diverse substituents are suitable substrates, affording the corresponding products in good yields and with complete diastereoselectivity. Additionally, the products were further converted to chlorinated products in good yields. More chemical transformations of borylated allylaziridines were also described.

https://doi.org/10.1021/acs.orglett.1c01372
Org. Lett. 2021, 23, 4431−4435

■ ASSOCIATED CONTENT

Supporting Information

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Experimental procedure, characterization data, NMR spectra, and X-ray crystallographic data (PDF)

Accession Codes

CCDC 2058388 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Support from the Natural Science Foundation of China (no. 22071265) is gratefully acknowledged.

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(14) For copies of the NMR spectrum for product **4aa**, please see the Supporting Information.

(15) For more information about screening conditions for the asymmetric version of the reaction, please see the Supporting Information.

https://doi.org/10.1021/acs.orglett.1c01372
Org. Lett. 2021, 23, 4431−4435